

Zero Coordinate Weights Break the Weak-Polar Implication and a Radial Positivity Hypothesis Repairs It

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Status. Candidate full counterexample and corrected result, likely valid, for the statements surrounding Theorem 1.3 of the source paper.

1 Source statement and definition

Koskela and Nguyen define a weak polar coordinate system $(\mathbb{S}, \sigma, \Gamma^O, h)$ by requiring a probability measure σ , a family of infinite curves γ_ξ^O starting from O , a weight $h : X \rightarrow [0, \infty)$, and a constant $C > 0$ such that

$$\int_{\mathbb{S}} \int_{\gamma_\xi^O} |f| h \, ds \, d\sigma(\xi) \leq C \int_X |f| \, d\mu \quad (1)$$

for every integrable f [1]. They explicitly note that h is not assumed strictly positive.

For $A_{2^j}(O) = B(O, 2^{j+1}) \setminus B(O, 2^j)$, they set

$$\begin{aligned} R_p(h, O) &= \int_{X \setminus B(O, 1)} h^{-p/(p-1)} \, d\mu, & p > 1, \\ \mathcal{R}_p(O) &= \sum_{j \in \mathbb{N}} (2^j)^{p/(p-1)} \mu(A_{2^j}(O))^{-1/(p-1)}, & p > 1, \end{aligned}$$

with the corresponding $p = 1$ definitions

$$R_1(h, O) = \|h^{-1}\|_{L_\mu^\infty(X \setminus B(O, 1))}, \quad \mathcal{R}_1(O) = \sup_j \frac{2^j}{\mu(A_{2^j}(O))}.$$

Theorem 1.3(I) states that $R_p(h, O) < \infty$ suffices for five radial-limit and modulus properties. Immediately afterward the authors infer $R_p(h, O) < \infty \Rightarrow \mathcal{R}_p(O) < \infty$ and ask for a simple direct proof [1, p. 3].

4. $\text{Mod}_p(\Gamma^O(\mathbb{S})) > 0$.
5. If $F \subseteq \mathbb{S}$ satisfies $\sigma(F) > 0$, then $\text{Mod}_p(\hat{\Gamma}^O(F)) > 0$.

We show that each of $\{1., 2., 3., 4., 5.\}$ is “between” $\mathcal{R}_p(O) < \infty$ and $R_p(h, O) < \infty$.

Theorem 1.3. *Let $1 \leq p < \infty$. Suppose that (X, d, μ) is a doubling metric measure space that supports a p -Poincaré inequality. Assume that X has the annular chain property at O . Suppose that X has a weak polar coordinate system $(\mathbb{S}, \sigma, \Gamma^O, h)$ at O as in (1.2). Then*

- I. *The condition $R_p(h, O) < \infty$ is sufficient for all of $\{1., 2., 3., 4., 5.\}$.*
- II. *The condition $\mathcal{R}_p(O) < \infty$ is necessary for each of $\{1., 2., 3., 4., 5.\}$.*

Theorem 1.3 leads us to compare the finiteness of $R_p(h, O)$ and $\mathcal{R}_p(O)$. First of all, it immediately follows from Theorem 1.3 that the finiteness of $R_p(h, O)$ guarantees the finiteness of $\mathcal{R}_p(O)$. We do not know a simple direct proof for this. On the other hand, the finiteness of $\mathcal{R}_p(O)$ does not in general yield the finiteness of $R_p(h, O)$. To see this, simply consider $\mathbb{R}^n$, $n \geq 2$, with the Euclidean distance and Lebesgue measure, usual spherical coordinates (with normalized measure on S^{n-1}) but with the coordinate weight $h = \chi_{\mathbb{R}^n \setminus B(0,1)}$. On the other hand, for example, if

$$h(x) \gtrsim \sum_{j \in \mathbb{N}} \frac{\mu(A_{2^j}(O))}{(2^j)^p} \chi_{A_{2^j}(O)}(x)$$

for all $x \in X$, then $R_p(h, O) \lesssim \mathcal{R}_p(O)$.

Let us next consider the Muckenhoupt $\mathcal{A}_p$ -weighted space $(\mathbb{R}^n, d_E, w)$, where $n \geq 2$ and O is the origin.

The decisive point in the printed definition is visible on source page 5:

A locally rectifiable curve γ is an *infinite curve* if $\gamma \setminus B \neq \emptyset$ for all balls B . Then $\int_\gamma ds = \infty$. We denote by Γ^∞ the collection of all infinite curves.

Let $\mathbb{S}$ be a nonempty set (a set of indices). Given a point O , we consider collections $\Gamma^O(\mathbb{S})$ of infinite curves with parameter space $\mathbb{S}$ starting from O , namely

$$\Gamma^O(\mathbb{S}) = \{\gamma_\xi^O \in \Gamma^\infty : \gamma_\xi^O(0) = O, \xi \in \mathbb{S}\}.$$

We say that (X, d, μ) has a *weak polar coordinate system* at the *coordinate point* O if there is a choice of a pair $(\mathbb{S}, \Gamma^O(\mathbb{S}))$ with a Radon probability measure σ on $\mathbb{S}$, a *coordinate weight* $h : X \rightarrow [0, \infty)$, and a constant $\mathcal{C} > 0$ such that

$$(2.1) \quad \int_{\mathbb{S}} \int_{\gamma_\xi^O} |f| h \, ds d\sigma \leq \mathcal{C} \int_X |f| d\mu \quad \text{for every integrable function } f.$$

Each $\gamma_\xi^O \in \Gamma^O(\mathbb{S})$ is called a *radial curve* with respect to $\xi \in \mathbb{S}$ (starting from O). Notice that we have not assumed h to be strictly positive. In applications it will be important to require h to be strictly positive at least almost everywhere in the union of the images of γ_ξ^O .

2 Proof intuition

The inequality (1) only sees the weighted line measure $h \, ds$. If h vanishes on every selected radial curve, the left side is identically zero and the condition places no geometric restriction on the family. The negative-power integral R_p , however, is computed using μ and ignores an infinite value on a μ -null ray. This mismatch gives a counterexample.

The natural repair is exactly the positivity mentioned after the source definition. Once $h > 0$ along almost every radial point, test (1) with a truncated version of $\mathbf{1}_A/h$. It recovers unweighted curve length. Every radial curve must spend length at least 2^j crossing the j th dyadic annulus, and one application of Hölder gives the corresponding summand of $\mathcal{R}_p$.

3 Counterexample

Theorem 1. *Under the printed definition of weak polar coordinates,*

$$R_p(h, O) < \infty \implies \mathcal{R}_p(O) < \infty$$

is false. Moreover, Theorem 1.3(I) of the source is false as printed.

Proof. Take

$$X = \mathbb{R}^2, \quad d = d_E, \quad d\mu = dx, \quad O = 0, \quad p = 2.$$

This is a doubling metric measure space supporting a 2-Poincaré inequality and the annular chain property. Let $\mathbb{S} = \{*\}$ with its unit point mass and choose the single radial curve

$$\gamma_*^O(t) = (t, 0), \quad t \geq 0.$$

Writing $L = \{(t, 0) : t \geq 0\}$, define the Borel coordinate weight

$$h(x) = \begin{cases} 0, & x \in L, \\ e^{|x|}, & x \notin L. \end{cases}$$

For every integrable f , the left side of (1) is

$$\int_0^\infty |f(t, 0)| h(t, 0) dt = 0.$$

Thus this is a weak polar coordinate system, even with $C = 1$.

The ray L is Lebesgue-null, so $h^{-2} = e^{-2|x|}$ almost everywhere. Consequently

$$R_2(h, 0) = \int_{\mathbb{R}^2 \setminus B(0,1)} h^{-2} dx = 2\pi \int_1^\infty r e^{-2r} dr = \frac{3\pi}{2} e^{-2} < \infty.$$

On the other hand,

$$\mu(A_{2^j}(0)) = \pi((2^{j+1})^2 - (2^j)^2) = 3\pi 4^j,$$

and hence

$$\mathcal{R}_2(0) = \sum_{j \in \mathbb{N}} \frac{(2^j)^2}{\mu(A_{2^j}(0))} = \sum_{j \in \mathbb{N}} \frac{1}{3\pi} = \infty.$$

This disproves the asserted implication.

It also directly contradicts Theorem 1.3(I). The tail family $\hat{\Gamma}^O(\mathbb{S})$ consists of one shifted ray. The indicator of its first unit segment is admissible for its 2-modulus, has line integral one, and has zero planar L^2 norm. Therefore

$$\text{Mod}_2(\hat{\Gamma}^O(\mathbb{S})) = 0,$$

whereas property 4 in the source theorem asserts strict positivity. □

4 Corrected direct theorem

The counterexample exploits only the omitted positivity condition. The following statement supplies both a repair and the requested elementary proof.

Proposition 1. *Assume the source's weak polar-coordinate inequality (1), and in addition assume*

$$h(\gamma_\xi^O(t)) > 0 \quad \text{for } ds \, d\sigma(\xi)\text{-almost every radial point.}$$

Then, whenever the right side is finite,

$$\mathcal{R}_p(O) \leq C^{p/(p-1)} R_p(h, O) \quad (p > 1), \quad \mathcal{R}_1(O) \leq C R_1(h, O).$$

In particular, $R_p(h, O) < \infty$ directly implies $\mathcal{R}_p(O) < \infty$. No doubling, Poincaré, or annular-chain hypothesis is used in this numerical implication beyond the standing finiteness of the annular measures.

Proof. Fix j and abbreviate $A_j = A_{2^j}(O)$. Every infinite continuous curve from O reaches radii 2^j and 2^{j+1} . Taking the last visit to the inner sphere before a first subsequent visit to the outer sphere shows that the intervening subcurve lies in A_j apart from its endpoints and has length at least 2^j . Thus

$$\int_{\gamma_\xi^O} \mathbf{1}_{A_j} \, ds \geq 2^j \quad (\xi \in \mathbb{S}). \quad (2)$$

Apply (1) to the bounded integrable functions

$$f_N = \mathbf{1}_{A_j} \min\{N, h^{-1}\}.$$

Radial almost-everywhere positivity and monotone convergence give

$$2^j \leq \int_{\mathbb{S}} \int_{\gamma_\xi^O} \mathbf{1}_{A_j} \, ds \, d\sigma \leq C \int_{A_j} h^{-1} \, d\mu.$$

For $p > 1$, write $q = p/(p-1)$. Hölder's inequality yields

$$\int_{A_j} h^{-1} \, d\mu \leq \mu(A_j)^{1/p} \left(\int_{A_j} h^{-q} \, d\mu \right)^{1/q}.$$

Raising to the power q and rearranging,

$$(2^j)^q \mu(A_j)^{-1/(p-1)} \leq C^q \int_{A_j} h^{-q} \, d\mu.$$

The annuli are disjoint, so summation over j gives $\mathcal{R}_p(O) \leq C^q R_p(h, O)$.

For $p = 1$, the same crossing argument gives

$$2^j \leq C \int_{A_j} h^{-1} \, d\mu \leq C R_1(h, O) \mu(A_j).$$

Taking the supremum over j proves the endpoint estimate. □

5 Scope, novelty, and verification

Theorem 1 concerns the literal published definition and statement. Proposition 1 does not assert that radial positivity is the weakest possible repair, but it is natural, is already flagged as important immediately after the source definition, and exactly removes the zero-weight mechanism.

A bounded search on 2026-08-17 covered arXiv:2203.01020, the 2023 accepted and journal versions, the exact phrase “do not know a simple direct proof”, the theorem title and authors together

with “erratum” and “correction”, and weak-polar/coordinate-weight keywords. The accepted version retains the same definition and request; no erratum, later explicit direct proof, or record of this zero-weight counterexample was found. This is not a priority claim. Novelty confidence is moderate and mathematical-validity confidence is high.

The exact integral and annular-term calculations are checked by the accompanying verification script; it is only a sanity check. The independent audit is in `VERIFICATION.md`. Human review should focus on the literal absence of radial positivity; the source itself explicitly emphasizes that strict positivity was not assumed.

References

- [1] P. Koskela and K. Nguyen, *Existence and uniqueness of limits at infinity for homogeneous Sobolev functions*, J. Funct. Anal. 285 (2023), no. 11, 110154; arXiv:2203.01020.